

Foundation (Jalapeno)

- Q1.** What is the slope of a line parallel to $y = 2x + 3$?
(A) $m = 2$
(B) $m = -2$
(C) $m = 1$
(D) $m = -1$
(E) $m = 0$
- Q2.** Which equation represents a line parallel to $y = 3x - 2$?
(A) $y = 3x + 2$
(B) $y = -3x + 2$
(C) $y = 2x - 3$
(D) $y = 2x + 3$
(E) $y = x - 2$
- Q3.** If two lines have the same slope, what can be said about them?
(A) They are perpendicular
(B) They are parallel
(C) They are skew
(D) They intersect at a 90° angle
(E) They are coincident
- Q4.** Which of the following is NOT a characteristic of parallel lines?
(A) Same slope
(B) Different y-intercepts
(C) Never intersect
(D) Perpendicular to the same line
(E) Intersect at a 90° angle
- Q5.** If $y = 2x + 1$ and $y = 2x - 3$, what is the relationship between these lines?
(A) Perpendicular
(B) Parallel
(C) Skew
(D) Coincident
(E) Intersecting
- Q6.** What is the equation of a line parallel to $y = x - 2$ that passes through $(1, 3)$?
(A) $y = x + 2$
(B) $y = x - 2$
(C) $y = x + 1$
(D) $y = x - 1$
(E) $y = 2x - 1$
- Q7.** Which lines are parallel: $y = 3x + 2$ and $y = -3x - 2$?
(A) Yes
(B) No
(C) Maybe
(D) Cannot be determined
(E) Only if they intersect
- Q8.** If $y = mx + b$ and $y = mx + c$, what can be said about these lines?
(A) They are perpendicular
(B) They are parallel
(C) They are skew
(D) They intersect at a 90° angle
(E) They are coincident

Open Ended Questions (FRQ)

Q9. Show that the lines $y = 2x + 1$ and $y = 2x - 3$ are parallel by finding their slopes.

Q10. Find the equation of a line parallel to $y = x + 2$ that passes through the point $(2, 5)$.

Chef's Tips

- Tip 1: Ensure students understand that parallel lines have the same slope.
- Tip 2: Emphasize that parallel lines never intersect.
- Tip 3: Remind students to check their work and use the given information to solve problems.